

COSC 120: Introduction to Python Programming

In-Class Midterm Examination

Prof. Jordan C. Hanson

Name: _____ Student ID:

Instructions: Choose the best answer for each question. Fill in the circle next to exactly one answer, A–D. Each question is worth 4 points.

Answer Sheet

1 ☐ A ☐ B ☐ C ☐ D	2 ☐ A ☐ B ☐ C ☐ D	3 ☐ A ☐ B ☐ C ☐ D	4 ☐ A ☐ B ☐ C ☐ D	5 ☐ A ☐ B ☐ C ☐ D
6 ☐ A ☐ B ☐ C ☐ D	7 ☐ A ☐ B ☐ C ☐ D	8 ☐ A ☐ B ☐ C ☐ D	9 ☐ A ☐ B ☐ C ☐ D	10 ☐ A ☐ B ☐ C ☐ D
11 ☐ A ☐ B ☐ C ☐ D	12 ☐ A ☐ B ☐ C ☐ D	13 ☐ A ☐ B ☐ C ☐ D	14 ☐ A ☐ B ☐ C ☐ D	15 ☐ A ☐ B ☐ C ☐ D
16 ☐ A ☐ B ☐ C ☐ D	17 ☐ A ☐ B ☐ C ☐ D	18 ☐ A ☐ B ☐ C ☐ D	19 ☐ A ☐ B ☐ C ☐ D	20 ☐ A ☐ B ☐ C ☐ D
21 ☐ A ☐ B ☐ C ☐ D	22 ☐ A ☐ B ☐ C ☐ D	23 ☐ A ☐ B ☐ C ☐ D	24 ☐ A ☐ B ☐ C ☐ D	25 ☐ A ☐ B ☐ C ☐ D

Python Fundamentals

1. What is printed by the following code?

```
x = 7
y = 3
print(x // y, x % y)
```

- ☐ A. 2 1
- ☐ B. 2 3
- ☐ C. 2.33 1
- ☐ D. 3 1

2. What is printed?

```
x = 3.14159
print(f"{x:.2f}")
```

- ☐ A. 3.1
- ☐ B. 3.14
- ☐ C. 3.142
- ☐ D. 3.14159

3. Consider

```
student = ("Maya", 20681001, 5)
```

What is the value of

```
student[2]
```

- ☐ A. Maya
- ☐ B. 20681001
- ☐ C. 5
- ☐ D. An error occurs

4. What is the final value of `numbers`?

```
numbers = [1, 2]
numbers.append(3)
```

- ☐ A. [1, 2]
- ☐ B. [1, 2, 3]
- ☐ C. [3, 1, 2]
- ☐ D. 1, 2, 3

5. What decimal value is produced by

```
int("1F", 16)
```

- ☐ A. 15
- ☐ B. 16
- ☐ C. 31
- ☐ D. 32

Boolean Logic and Decisions

6. Consider the placement code

```
score = 3

if score >= 4:
    course = "Calculus"
elif score >= 2:
    course = "Trigonometry"
else:
    course = "Algebra"
```

What is stored in `course`?

- ☐ A. Calculus
- ☐ B. Trigonometry
- ☐ C. Algebra
- ☐ D. Nothing

7. What is the value of

`True or False and False`

Recall that `and` is evaluated before `or`.

- ☐ A. True
- ☐ B. False
- ☐ C. 0
- ☐ D. An error occurs

8. A screening function contains

```
age > 70 and
weight > 200 and
gender == "M"
```

Which patient satisfies the condition?

- ☐ A. Male, age 70, weight 210
- ☐ B. Male, age 75, weight 195
- ☐ C. Female, age 75, weight 220
- ☐ D. Male, age 75, weight 220

Loops and Nested Lists

9. What values are printed?

```
for i in range(1, 4):
    print(i)
```

- ☐ A. 0, 1, 2, 3
- ☐ B. 1, 2, 3
- ☐ C. 1, 2, 3, 4
- ☐ D. 0, 1, 2

10. How many times does this loop execute?

```
i = 0

while i < 3:
    print(i)
    i = i + 1
```

- ☐ A. 2
- ☐ B. 3
- ☐ C. 4
- ☐ D. The loop never stops

11. Consider

```
for i in range(len(courses)):

    for j in range(i + 1,
                    len(courses)):
        ...
```

Why does the inner loop begin at `i + 1`?

- ☐ A. To skip every other course
- ☐ B. To compare each course only with itself
- ☐ C. To avoid comparing a pair twice and avoid comparing a course with itself
- ☐ D. Because Python lists begin at index 1

12. Consider the matrix

```
A = [
    [4, 8, 2],
    [7, 5, 9],
    [1, 6, 3]
]
```

What is the value of

`A[1][2]`

- ☐ A. 5
- ☐ B. 7
- ☐ C. 9
- ☐ D. 6

Functions

13. Consider

```
def square(x):
    return x * x

square(5)
```

Which statement is correct?

- ☐ A. `x` is the argument and 5 is the parameter
- ☐ B. `x` is the parameter and 5 is the argument
- ☐ C. Both `x` and 5 are parameters
- ☐ D. Neither is a parameter

14. Why is `return` useful in a function such as

```
def find_patient(...):
    ...
    return patient
```

- ☐ A. It permanently prints the patient
- ☐ B. It sends a value back to the calling code
- ☐ C. It automatically creates a loop
- ☐ D. It imports another module

15. In our Fourier-series exercise, the functions were organized as

```
plot_series()
sawtooth()
fourier_term()
```

Which call sequence correctly describes the program?

- ☐ A. `fourier_term()` → `sawtooth()` → `plot_series()`
- ☐ B. `plot_series()` → `sawtooth()` → `fourier_term()`
- ☐ C. `sawtooth()` → `plot_series()` → `fourier_term()`
- ☐ D. None of the functions call one another

Modules

16. Suppose `math_tools.py` contains a function called `square()`. Which code correctly imports the module and calls the function?

- ☐ A. `import math_tools`
`math_tools.square(5)`

- ☐ B. `import square`
`math_tools(5)`
- ☐ C. `open math_tools`
`square.math_tools(5)`
- ☐ D. `module math_tools`
`square(5)`

17. Which code correctly imports only the function `square()`?]

- ☐ A. `import square from math_tools`
- ☐ B. `from math_tools import square`
- ☐ C. `from square import math_tools`
- ☐ D. `math_tools import(square)`

18. Before a custom file such as

`course_tools.py`

can be imported into a Google Colab session, what must normally be done?

- ☐ A. Convert it to a PDF
- ☐ B. Upload the Python file to the Colab runtime
- ☐ C. Rename it with a `.txt` extension
- ☐ D. Place all functions inside `print()`

Strings and Searching

19. What is the value of

`"Garcia" in "Maria Garcia"`

- ☐ A. True
- ☐ B. False
- ☐ C. Garcia
- ☐ D. An error occurs

20. Which condition performs a case-insensitive partial-name search?

- ☐ A. `search == patient[1]`
- ☐ B. `search in patient[1]`
- ☐ C. `search.lower() in patient[1].lower()`
- ☐ D. `search.upper() == patient[1].lower()`

21. A function searches every patient record but finds no matching patient ID. What value did our exercise specify should be returned?

- ☐ A. 0
- ☐ B. False
- ☐ C. `""`
- ☐ D. None

Course Lists and Sorting

22. A course is stored as

```
course = [
    "PHYS", 100, "SLC", 202,
    "MWF", 1400
```

What is the value of `course[5]`?

- ☐ A. SLC
- ☐ B. 202
- ☐ C. MWF
- ☐ D. 1400

23. Suppose

```
def get_time(course):
    return course[5]

def sort_by_time(courses):
    courses.sort(key=get_time)
```

What does `key=get_time` tell Python to do?

- ☐ A. Sort using the course meeting time
- ☐ B. Delete the meeting time
- ☐ C. Sort only courses named "Time"
- ☐ D. Replace each course with its meeting time

24. Which time would appear first after sorting numerically?

930, 1400, 1100, 1530

- ☐ A. 930
- ☐ B. 1100
- ☐ C. 1400
- ☐ D. 1530

25. Consider these two courses:

```
["PHYS", 100, "SLC", 202,
 "MWF", 1400]
```

```
["MATH", 200, "SLC", 202,
 "TR", 1400]
```

Using the room-conflict rule from our course-schedule exercise, do these courses conflict?

- ☐ **A.** Yes, because the room is the same
- ☐ **B.** Yes, because the time is the same
- ☐ **C.** No, because they do not share a meeting day
- ☐ **D.** No, because the course numbers are different

End of Exam

Check that exactly one answer is clearly marked for each question.